Chapter 5: Problems with solutions

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2. If a man weighs 900 N on the Earth, what would he weigh on Jupiter, where the free-fall acceleration is 25.9 m/s^2?

We are given $F_g = mg = 900\ N$, from which we can find the man's mass,

$$m = \frac{900\ N}{9.80\ m/s^2} = 91.8\ kg$$

Then, his weight on Jupiter is given by

$$(F_g)_{\text{on Jupiter}} = 91.8\ kg(25.9\ m/s^2) = \boxed{2.38\ kN}$$

3. **W** A 3.00-kg object undergoes an acceleration given by $\vec{a} = (2.00\hat{i} + 5.00\hat{j})\ \text{m/s}^2$. Find (a) the resultant force acting on the object and (b) the magnitude of the resultant force.

We use Newton's second law to find the force as a vector and then the Pythagorean theorem to find its magnitude. The givens are $m = 3.00$ kg and $\vec{a} = \left(2.00\hat{i} + 5.00\hat{j}\right)\ \text{m/s}^2$.

(a) The total vector force is

$$\Sigma\vec{F} = m\vec{a} = (3.00\ \text{kg})(2.00\hat{i} + 5.00\hat{j})\ \text{m/s}^2 = \boxed{(6.00\hat{i} + 15.0\hat{j})\ \text{N}}$$

(b) Its magnitude is

$$\left|\vec{F}\right| = \sqrt{\left(F_x\right)^2 + \left(F_y\right)^2} = \sqrt{(6.00\ \text{N})^2 + (15.0\ \text{N})^2} = \boxed{16.2\ \text{N}}$$

4. A certain orthodontist uses a wire brace to align a patient's crooked tooth as in Figure P5.4. The tension in the wire is adjusted to have a magnitude of 18.0 N. Find the magnitude of the net force exerted by the wire on the crooked tooth.

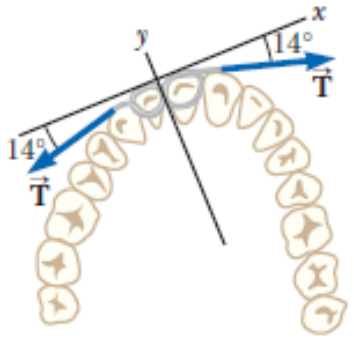


Figure P5.4

Using the reference axes shown in Figure P5.4, we see that

$$\sum F_x = T\cos 14.0^\circ - T\cos 14.0^\circ = 0$$

and

$$\sum F_y = -T\sin 14.0^\circ - T\sin 14.0^\circ = -2T\sin 14.0^\circ$$

Thus, the magnitude of the resultant force exerted on the tooth by the wire brace is

$$R = \sqrt{\left(\sum F_x\right)^2 + \left(\sum F_y\right)^2} = \sqrt{0 + (-2T\sin 14.0^\circ)^2} = 2T\sin 14.0^\circ$$

or

$$R = 2(18.0\text{ N})\sin 14.0^\circ = \boxed{8.71\text{ N}}$$

5. **M** A toy rocket engine is securely fastened to a large puck that can glide with negligible friction over a horizontal surface, taken as the xy plane. The 4.00-kg puck has a velocity of $3.00\hat{\mathbf{i}}$ m/s at one instant. Eight seconds later, its velocity is $(8.00\hat{\mathbf{i}} + 10.00\hat{\mathbf{j}})$ m/s. Assuming the rocket engine exerts a constant horizontal force, find (a) the components of the force and (b) its magnitude.

We use the particle under constant acceleration and particle under a net force models. We first calculate the acceleration of the puck:

$$\vec{\mathbf{a}} = \frac{\Delta\vec{\mathbf{v}}}{\Delta t} = \frac{\left(8.00\hat{\mathbf{i}} + 10.0\hat{\mathbf{j}}\right)\text{m/s} - 3.00\hat{\mathbf{i}}\ \text{m/s}}{8.00\ \text{s}}$$

$$= 0.625\hat{\mathbf{i}}\ \text{m/s}^2 + 1.25\hat{\mathbf{j}}\ \text{m/s}^2$$

In $\sum\vec{\mathbf{F}} = m\vec{\mathbf{a}}$, the only horizontal force is the thrust $\vec{\mathbf{F}}$ of the rocket:

(a) $$\vec{\mathbf{F}} = (4.00\ \text{kg})\left(0.625\hat{\mathbf{i}}\ \text{m/s}^2 + 1.25\hat{\mathbf{j}}\ \text{m/s}^2\right) = \boxed{\left(2.50\hat{\mathbf{i}} + 5.00\hat{\mathbf{j}}\right)\ \text{N}}$$

(b) Its magnitude is $$|\vec{\mathbf{F}}| = \sqrt{(2.50\ \text{N})^2 + (5.00\ \text{N})^2} = \boxed{5.59\ \text{N}}$$

6. The average speed of a nitrogen molecule in air is about 6.70×10^2 m/s, and its mass is 4.68×10^{-26} kg. (a) If it takes 3.00×10^{-13} s for a nitrogen molecule to hit a wall and rebound with the same speed but moving in the opposite direction, what is the average acceleration of the molecule during this time interval? (b) What average force does the molecule exert on the wall?

(a) Let the x axis be in the original direction of the molecule's motion. Then, from $v_f = v_i + at$, we have

$$a = \frac{v_f - v_i}{t} = \frac{-670\ \text{m/s} - 670\ \text{m/s}}{3.00 \times 10^{-13}\ \text{s}} = \boxed{-4.47 \times 10^{15}\ \text{m/s}^2}$$

(b) For the molecule, $\sum \vec{F} = m\vec{a}$. Its weight is negligible.

$$\vec{F}_{\text{wall on molecule}} = \left(4.68 \times 10^{-26}\ \text{kg}\right)\left(-4.47 \times 10^{15}\ \text{m/s}^2\right)$$

$$= -2.09 \times 10^{-10}\ \text{N}$$

$$\vec{F}_{\text{molecule on wall}} = \boxed{+2.09 \times 10^{-10}\ \text{N}}$$

8. (a) A car with a mass of 850 kg is moving to the right with a constant speed of 1.44 m/s. What is the total force on the car? (b) What is the total force on the car if it is moving to the left?

The force on the car is given by $\sum \vec{F} = m\vec{a}$, or, in one dimension, $\sum F = ma$. Whether the car is moving to the left or the right, since it's moving at constant speed, $a = 0$ and therefore $\sum F = \boxed{0}$ for both parts (a) and (b).

11. **Review.** An electron of mass 9.11×10^{-31} kg has an **M** initial speed of 3.00×10^5 m/s. It travels in a straight line, and its speed increases to 7.00×10^5 m/s in a distance of 5.00 cm. Assuming its acceleration is constant, (a) determine the magnitude of the force exerted on the electron and (b) compare this force with the weight of the electron, which we ignored.

(a) From $v_f^2 = v_i^2 + 2ax$ and $\sum F = ma$, we can solve for the acceleration and then the force: $a = \dfrac{v_f^2 - v_i^2}{2x}$

Substituting to eliminate a, $\sum F = \dfrac{m\left(v_f^2 - v_i^2\right)}{2x}$

Substituting the given information,

$$\sum F = \frac{\left(9.11 \times 10^{-31}\ \text{kg}\right)\left[\left(7.00 \times 10^5\ \text{m/s}\right)^2 - \left(3.00 \times 10^5\ \text{m/s}\right)^2\right]}{2(0.050\ 0\ \text{m})}$$

$$\sum F = \boxed{3.64 \times 10^{-18}\ \text{N}}$$

(b) The Earth exerts on the electron the force called weight,

$$F_g = mg = (9.11 \times 10^{-31}\,\text{kg})(9.80\ \text{m/s}^2) = 8.93 \times 10^{-30}\,\text{N}$$

The accelerating force is

$\boxed{4.08 \times 10^{11}\ \text{times the weight of the electron.}}$

12. Besides the gravitational force, a 2.80-kg object is subjected to one other constant force. The object starts from rest and in 1.20 s experiences a displacement of $(4.20\hat{\mathbf{i}} - 3.30\hat{\mathbf{j}})$ m, where the direction of $\hat{\mathbf{j}}$ is the upward vertical direction. Determine the other force.

We first find the acceleration of the object:

$$\vec{r}_f - \vec{r}_i = \vec{v}_i t + \frac{1}{2}\vec{a}t^2$$

$$4.20\ \text{m}\hat{\mathbf{i}} - 3.30\ \text{m}\hat{\mathbf{j}} = 0 + \frac{1}{2}\vec{a}(1.20\ \text{s})^2 = (0.720\ \text{s}^2)\vec{a}$$

$$\vec{a} = \left(5.83\hat{\mathbf{i}} - 4.58\hat{\mathbf{j}}\right)\ \text{m/s}^2$$

Now $\sum \vec{F} = m\vec{a}$ becomes

$$\vec{F}_g + \vec{F}_2 = m\vec{a}$$

$$\vec{F}_2 = 2.80\ \text{kg}\left(5.83\hat{\mathbf{i}} - 4.58\hat{\mathbf{j}}\right)\ \text{m/s}^2 + (2.80\ \text{kg})(9.80\ \text{m/s}^2)\hat{\mathbf{j}}$$

$$\vec{F}_2 = \boxed{\left(16.3\hat{\mathbf{i}} + 14.6\hat{\mathbf{j}}\right)\ \text{N}}$$

15. Two forces, $\vec{\mathbf{F}}_1 = (-6.00\hat{\mathbf{i}} - 4.00\hat{\mathbf{j}})$ N and $\vec{\mathbf{F}}_2 = (-3.00\hat{\mathbf{i}} + 7.00\hat{\mathbf{j}})$ N, act on a particle of mass 2.00 kg that is initially at rest at coordinates (−2.00 m, +4.00 m). (a) What are the components of the particle's velocity at $t = 10.0$ s? (b) In what direction is the particle moving at $t = 10.0$ s? (c) What displacement does the particle undergo during the first 10.0 s? (d) What are the coordinates of the particle at $t = 10.0$ s?

(a) We start from the sum of the two forces:

$$\sum \vec{\mathbf{F}} = \vec{\mathbf{F}}_1 + \vec{\mathbf{F}}_2 = \left(-6.00\hat{\mathbf{i}} - 4.00\hat{\mathbf{j}}\right) + \left(-3.00\hat{\mathbf{i}} + 7.00\hat{\mathbf{j}}\right)$$
$$= \left(-9.00\hat{\mathbf{i}} + 3.00\hat{\mathbf{j}}\right) \text{ N}$$

The acceleration is then:

$$\vec{\mathbf{a}} = a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}} = \frac{\sum \vec{\mathbf{F}}}{m} = \frac{\left(-9.00\hat{\mathbf{i}} + 3.00\hat{\mathbf{j}}\right) \text{ N}}{2.00 \text{ kg}}$$
$$= \left(-4.50\hat{\mathbf{i}} + 1.50\hat{\mathbf{j}}\right) \text{ m/s}^2$$

and the velocity is found from

$$\vec{\mathbf{v}}_f = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t = \vec{\mathbf{a}}t$$

$$\vec{\mathbf{v}}_f = \left[\left(-4.50\hat{\mathbf{i}} + 1.50\hat{\mathbf{j}}\right) \text{ m/s}^2\right](10.0 \text{ s})$$
$$= \boxed{\left(-45.0\hat{\mathbf{i}} + 15.0\hat{\mathbf{j}}\right) \text{ m/s}}$$

(b) The direction of motion makes angle θ with the x direction.

$$\theta = \tan^{-1}\left(\frac{v_y}{v_x}\right) = \tan^{-1}\left(-\frac{15.0\ \text{m/s}}{45.0\ \text{m/s}}\right)$$

$$\theta = -18.4^\circ + 180^\circ = \boxed{162^\circ \text{ from the } +x \text{ axis}}$$

(c) Displacement:

$$x\text{-displacement} = x_f - x_i = v_{xi}t + \frac{1}{2}a_x t^2$$

$$= \frac{1}{2}\left(-4.50\ \text{m/s}^2\right)(10.0\ \text{s})^2 = -225\ \text{m}$$

$$y\text{-displacement} = y_f - y_i = v_{yi}t + \frac{1}{2}a_y t^2$$

$$= \frac{1}{2}\left(+1.50\ \text{m/s}^2\right)(10.0\ \text{s})^2 = +75.0\ \text{m}$$

$$\Delta\vec{r} = \boxed{\left(-225\hat{i} + 75.0\hat{j}\right)\ \text{m}}$$

(d) Position: $\vec{r}_f = \vec{r}_i + \Delta\vec{r}$

$$\vec{r}_f = \left(-2.00\hat{i} + 4.00\hat{j}\right) + \left(-225\hat{i} + 75.0\hat{j}\right) = \boxed{\left(-227\hat{i} + 79.0\hat{j}\right)\ \text{m}}$$

16. M The force exerted by the wind on the sails of a sailboat is 390 N north. The water exerts a force of 180 N east. If the boat (including its crew) has a mass of 270 kg, what are the magnitude and direction of its acceleration?

Since the two forces are perpendicular to each other, their resultant is

$$F_R = \sqrt{(180\ \text{N})^2 + (390\ \text{N})^2} = 430\ \text{N}$$

at an angle of

$$\theta = \tan^{-1}\left(\frac{390\ \text{N}}{180\ \text{N}}\right) = 65.2^\circ\ \text{N of E}$$

From Newton's second law,

$$a = \frac{F_R}{m} = \frac{430\ \text{N}}{270\ \text{kg}} = 1.59\ \text{m/s}^2$$

or

$$\vec{a} = \boxed{1.59\ \text{m/s}^2\ \text{at}\ 65.2^\circ\ \text{N of E}}$$

18. A force $\vec{\mathbf{F}}$ applied to an object of mass m_1 produces an acceleration of 3.00 m/s^2. The same force applied to a second object of mass m_2 produces an acceleration of 1.00 m/s^2. (a) What is the value of the ratio m_1/m_2? (b) If m_1 and m_2 are combined into one object, find its acceleration under the action of the force $\vec{\mathbf{F}}$.

W

For the same force F, acting on different masses $F = m_1 a_1$ and $F = m_2 a_2$. Setting these expressions for F equal to one another gives:

(a) $$\frac{m_1}{m_2} = \frac{a_2}{a_1} = \boxed{\frac{1}{3}}$$

(b) The acceleration of the combined object is found from

$$F = (m_1 + m_2)a = 4m_1 a$$

or $$a = \frac{F}{4m_1} = \frac{1}{4}(3.00\ \text{m/s}^2) = \boxed{0.750\ \text{m/s}^2}$$

19. M Two forces $\vec{F}_1$ and $\vec{F}_2$ act on a 5.00-kg object. Taking $F_1 = 20.0$ N and $F_2 = 15.0$ N, find the accelerations of the object for the configurations of forces shown in parts (a) and (b) of Figure P5.19.

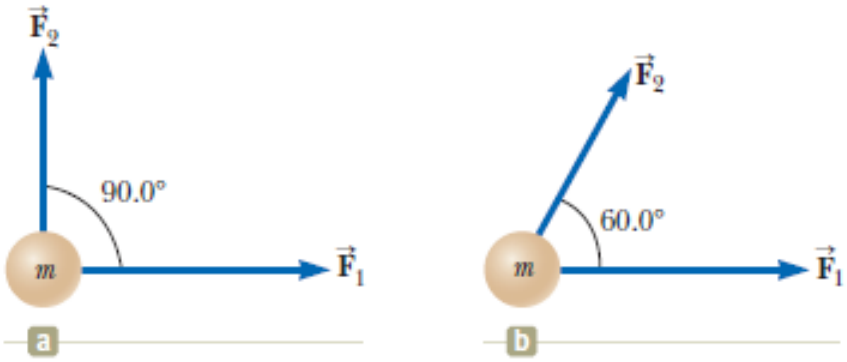


Figure P5.19

(a) $\sum \vec{F} = \vec{F}_1 + \vec{F}_2 = (20.0\hat{i} + 15.0\hat{j})$ N

Newton's second law gives, with $m = 5.00$ kg,

$$\vec{a} = \frac{\sum \vec{F}}{m} = \left(4.00\hat{i} + 3.00\hat{j}\right) \text{ m/s}^2$$

or $\boxed{a = 5.00 \text{ m/s}^2 \text{ at } \theta = 36.9°}$

(b) In this configuration,

$$F_{2x} = 15.0\cos 60.0° = 7.50 \text{ N}$$
$$F_{2y} = 15.0\sin 60.0° = 13.0 \text{ N}$$
$$\vec{F}_2 = \left(7.50\hat{i} + 13.0\hat{j}\right) \text{ N}$$

Then,

$$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 = \left[20.0\hat{i} + \left(7.50\hat{i} + 13.0\hat{j}\right)\right] \text{N}$$
$$= (27.5\hat{i} + 13.0\hat{j}) \text{ N}$$

$$\vec{a} = \frac{\sum \vec{F}}{m} = \left(5.50\hat{i} + 2.60\hat{j}\right) \text{ m/s}^2 = \boxed{6.08 \text{ m/s}^2 \text{ at } 25.3°}$$

22. **Review.** Three forces acting on an object are given by **W** $\vec{F}_1 = (-2.00\hat{i} + 2.00\hat{j})$ N, and $\vec{F}_2 = (5.00\hat{i} - 3.00\hat{j})$ N, and $\vec{F}_3 = (-45.0\hat{i})$ N. The object experiences an acceleration of magnitude 3.75 m/s^2. (a) What is the direction of the acceleration? (b) What is the mass of the object? (c) If the object is initially at rest, what is its speed after 10.0 s? (d) What are the velocity components of the object after 10.0 s?

$\sum \vec{F} = m\vec{a}$ reads

$$\left(-2.00\hat{i} + 2.00\hat{j} + 5.00\hat{i} - 3.00\hat{j} - 45.0\hat{i}\right) \text{ N} = m\left(3.75 \text{ m/s}^2\right)\hat{a}$$

where $\hat{a}$ represents the direction of $\vec{a}$:

$$\left(-42.0\hat{i} - 1.00\hat{j}\right) \text{ N} = m\left(3.75 \text{ m/s}^2\right)\hat{a}$$

$$\sum \vec{F} = \sqrt{(42.0)^2 + (1.00)^2} \text{ N at } \tan^{-1}\left(\frac{1.00}{42.0}\right) \text{ below the } -x \text{ axis}$$

$$\sum \vec{F} = 42.0 \text{ N at } 181^\circ = m\left(3.75 \text{ m/s}^2\right)\hat{a}$$

For the vectors to be equal, their magnitudes and their directions must be equal.

(a) Therefore $\boxed{\hat{a} \text{ is at } 181^\circ}$ counter-clockwise from the x axis

(b) $$m = \frac{42.0\ \text{N}}{3.75\ \text{m/s}^2} = \boxed{11.2\ \text{kg}}$$

(c) $$v = |\vec{v}| = 0 + |\vec{a}|t = (3.75\ \text{m/s}^2)(10.00\ \text{s}) = \boxed{37.5\ \text{m/s}}$$

(d) $$\vec{v} = \vec{v}_i + |\vec{a}|\,t = 0 + \frac{F}{m}t$$

$$\vec{v} = \frac{\left(-42.0\hat{i} - 1.00\hat{j}\right)\ \text{N}}{11.2\ \text{kg}}(10.0\ \text{s}) = \left(-37.5\hat{i} - 0.893\hat{j}\right)\ \text{m/s}$$

So, $\vec{v}_f = \boxed{\left(-37.5\hat{i} - 0.893\hat{j}\right)\ \text{m/s}}$

29. Assume the three blocks portrayed in Figure P5.29 **M** move on a frictionless surface and a 42-N force acts as shown on the 3.0-kg block. Determine (a) the acceleration given this system, (b) the tension in the cord connecting the 3.0-kg and the 1.0-kg blocks, and (c) the force exerted by the 1.0-kg block on the 2.0-kg block.

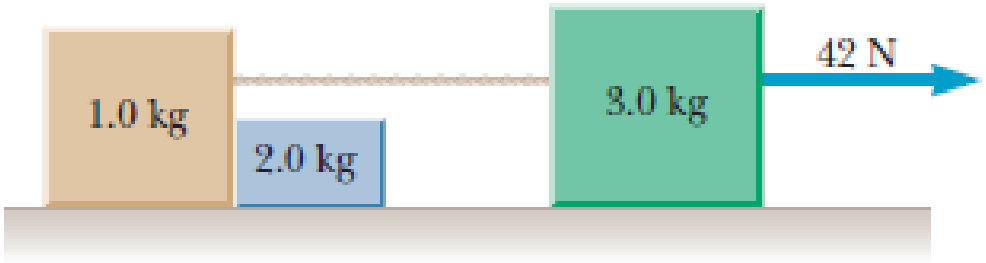


Figure P5.29

(a) The resultant external force acting on this system, consisting of all three blocks having a total mass of 6.0 kg, is 42 N directed horizontally toward the right. Thus, the acceleration produced is

$$a = \frac{\sum F}{m} = \frac{42\ \text{N}}{6.0\ \text{kg}} = \boxed{7.0\ \text{m/s}^2 \text{ horizontally to the right}}$$

(b) Draw a free-body diagram of the 3.0-kg block and apply Newton's second law to the horizontal forces acting on this block:

$$\sum F_x = ma_x:$$

$$42\ \text{N} - T = (3.0\ \text{kg})(7.0\ \text{m/s}^2) \quad \rightarrow \quad T = \boxed{21\ \text{N}}$$

(c) The force accelerating the 2.0-kg block is the force exerted on it by the 1.0-kg block. Therefore, this force is given by

$$F = ma = (2.0\ \text{kg})(7.0\ \text{m/s}^2) = 14\ \text{N}$$

or $\vec{F} =$ **14 N horizontally to the right**

32. A 3.00-kg object is moving in a plane, with its x and y coordinates given by $x = 5t^2 - 1$ and $y = 3t^3 + 2$, where x and y are in meters and t is in seconds. Find the magnitude of the net force acting on this object at $t = 2.00$ s.

To find the net force, we differentiate the equations for the position of the particle once with respect to time to obtain the velocity, and once again to obtain the acceleration:

$$v_x = \frac{dx}{dt} = \frac{d}{dt}\left(5t^2 - 1\right) = 10t, \quad v_y = \frac{dy}{dt} = \frac{d}{dt}\left(3t^3 + 2\right) = 9t^2$$

$$a_x = \frac{dv_x}{dt} = 10, \quad a_y = \frac{dv_y}{dt} = 18t$$

Then, at $t = 2.00$ s, $a_x = 10.0\ \text{m/s}^2$, $a_y = 36.0\ \text{m/s}^2$, and Newton's second law gives us

$$\sum F_x = ma_x: \quad 3.00\ \text{kg}(10.0\ \text{m/s}^2) = 30.0\ \text{N}$$

$$\sum F_y = ma_y: \quad 3.00\ \text{kg}(36.0\ \text{m/s}^2) = 108\ \text{N}$$

$$\sum F = \sqrt{F_x^2 + F_y^2} = \boxed{112\ \text{N}}$$

33. A bag of cement weighing 325 N hangs in equilibrium from three wires as suggested in Figure P5.33. Two of the wires make angles $\theta_1 = 60.0°$ and $\theta_2 = 40.0°$ with the horizontal. Assuming the system is in equilibrium, find the tensions T_1, T_2, and T_3 in the wires.

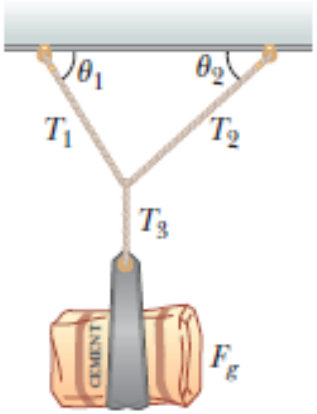


Figure P5.33

From equilibrium of the sack:

$$T_3 = F_g \qquad [1]$$

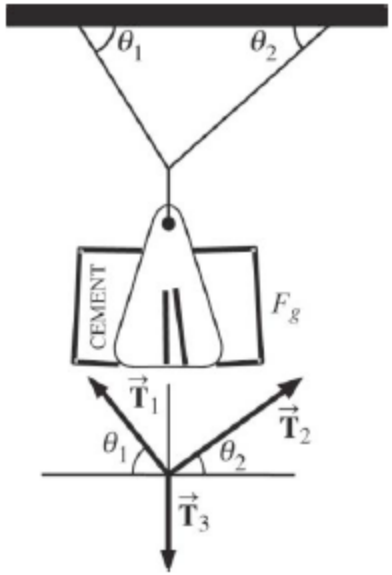
From $\sum F_y = 0$ for the knot:

$$T_1 \sin\theta_1 + T_2 \sin\theta_2 = F_g \qquad [2]$$

From $\sum F_x = 0$ for the knot:

$$T_1 \cos\theta_1 = T_2 \cos\theta_2 \qquad [3]$$

Eliminate T_2 by using $T_2 = T_1 \cos\theta_1 / \cos\theta_2$ and solve for T_1:



ANS. FIG. P5.33

$$T_1 = \frac{F_g \cos\theta_2}{(\sin\theta_1 \cos\theta_2 + \cos\theta_1 \sin\theta_2)} = \frac{F_g \cos\theta_2}{\sin(\theta_1 + \theta_2)}$$

$$T_3 = F_g = \boxed{325 \text{ N}}$$

$$T_1 = F_g\left(\frac{\cos 40.0°}{\sin 100.0°}\right) = \boxed{253 \text{ N}}$$

$$T_2 = T_1\left(\frac{\cos\theta_1}{\cos\theta_2}\right) = (253 \text{ N})\left(\frac{\cos 60.0°}{\cos 40.0°}\right) = \boxed{165 \text{ N}}$$

37. An object of mass $m = 1.00$ kg is observed to have an acceleration $\vec{a}$ with a magnitude of 10.0 m/s^2 in a direction 60.0° east of north. Figure P5.37 shows a view of the object from above. The force $\vec{F}_2$ acting on the object has a magnitude of 5.00 N and is directed north. Determine the magnitude and direction of the one other horizontal force $\vec{F}_1$ acting on the object.

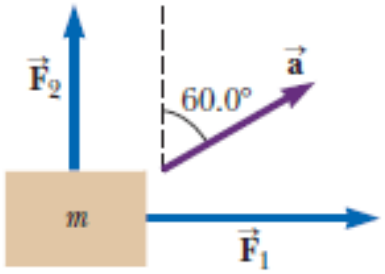
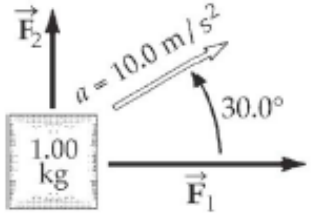


Figure P5.37

Choose a coordinate system with $\hat{i}$ East and $\hat{j}$ North. The acceleration is

$$\vec{a} = [(10.0 \cos 30.0°)\hat{i} + (10.0 \sin 30.0°)\hat{j}] \text{ m/s}^2$$
$$= (8.66\hat{i} + 5.00\hat{j}) \text{ m/s}^2$$



ANS. FIG. P5.37

From Newton's second law,

$$\sum \vec{F} = m\vec{a} = (1.00 \text{ kg})(8.66\hat{i} \text{ m/s}^2 + 5.00\hat{j} \text{ m/s}^2)$$
$$= (8.66\hat{i} + 5.00\hat{j}) \text{ N}$$

and $\sum \vec{F} = \vec{F}_1 + \vec{F}_2$

So the force we want is

$$\vec{F}_1 = \sum \vec{F} - \vec{F}_2 = (8.66\hat{i} + 5.00\hat{j} - 5.00\hat{j}) \text{ N}$$
$$= 8.66\hat{i} \text{ N} = \boxed{8.66 \text{ N east}}$$

43. Two blocks, each of mass $m = 3.50$ kg, are hung from the ceiling of an elevator as in Figure P5.43. (a) If the elevator moves with an upward acceleration $\vec{a}$ of magnitude 1.60 m/s^2, find the tensions T_1 and T_2 in the upper and lower strings. (b) If the strings can withstand a maximum tension of 85.0 N, what maximum acceleration can the elevator have before a string breaks?

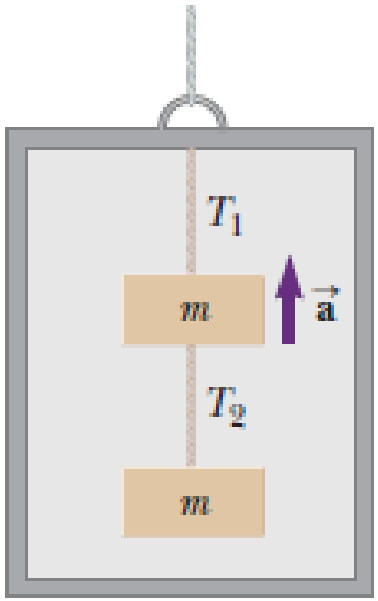
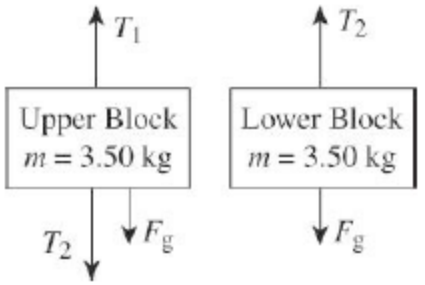


Figure P5.43

(a) Free-body diagrams of the two blocks are shown in ANS. FIG. P5.43. Note that each block experiences a downward gravitational force



ANS. FIG. P5.43

$$F_g = (3.50\ \text{kg})(9.80\ \text{m/s}^2) = 34.3\ \text{N}$$

Also, each has the same upward acceleration as the elevator, in this case $a_y = +1.60\ \text{m/s}^2$.

Applying Newton's second law to the lower block:

$$\sum F_y = ma_y \quad \Rightarrow \quad T_2 - F_g = ma_y$$

or

$$T_2 = F_g + ma_y = 34.3\ \text{N} + (3.50\ \text{kg})(1.60\ \text{m/s}^2) = \boxed{39.9\ \text{N}}$$

Next, applying Newton's second law to the upper block:

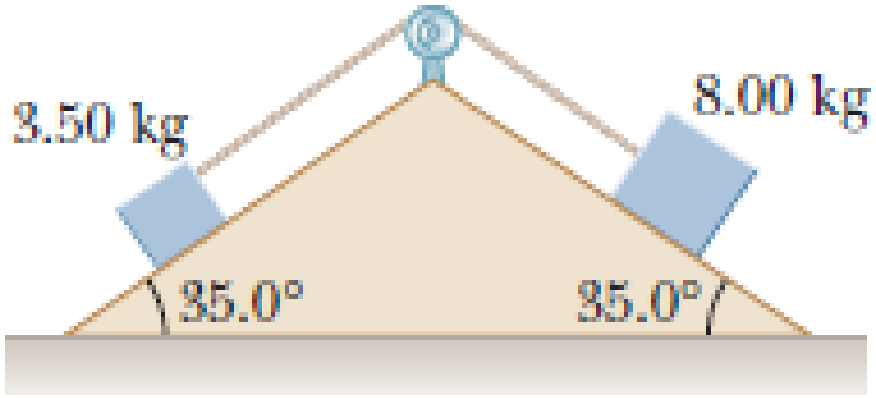
$$\sum F_y = ma_y \quad \Rightarrow \quad T_1 - T_2 - F_g = ma_y$$

or

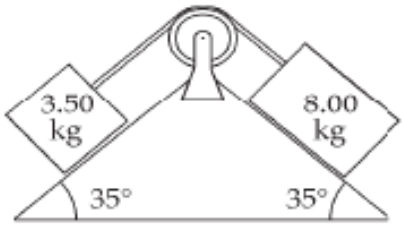
$$T_1 = T_2 + F_g + ma_y = 39.9\ \text{N} + 34.3\ \text{N} + (3.50\ \text{kg})(1.60\ \text{m/s}^2)$$

$$= \boxed{79.8\ \text{N}}$$

49. Two blocks of mass 3.50 kg and 8.00 kg are connected by a massless string that passes over a frictionless pulley (Fig. P5.49). The inclines are frictionless. Find (a) the magnitude of the acceleration of each block and (b) the tension in the string.



Since it has a larger mass, we expect the 8.00-kg block to move down the plane. The acceleration for both blocks should have the same magnitude since they are joined together by a non-stretching string. Define up the left-hand plane as positive for the 3.50-kg object and down the right-hand plane as positive for the 8.00-kg object.



ANS. FIG. P5.49

$$\sum F_1 = m_1 a_1: \quad -m_1 g \sin 35.0° + T = m_1 a$$
$$\sum F_2 = m_2 a_2: \quad m_2 g \sin 35.0° - T = m_2 a$$

and, suppressing units,

$$-(3.50)(9.80)\sin 35.0° + T = 3.50a$$
$$(8.00)(9.80)\sin 35.0° - T = 8.00a.$$

Adding, we obtain $+45.0\ \text{N} - 19.7\ \text{N} = (11.5\ \text{kg})a$.

(a) Thus the acceleration is $\boxed{a = 2.20\ \text{m/s}^2}$. By substitution,

$$-19.7\ \text{N} + T = (3.50\ \text{kg})(2.20\ \text{m/s}^2) = 7.70\ \text{N}$$

(b) The tension is $\boxed{T = 27.4\ \text{N}}$

65. AMT M Two blocks connected by a rope of negligible mass are being dragged by a horizontal force (Fig. P5.65). Suppose $F = 68.0$ N, $m_1 = 12.0$ kg, $m_2 = 18.0$ kg, and the coefficient of kinetic friction between each block and the surface is 0.100. (a) Draw a free-body diagram for each block. Determine (b) the acceleration of the system and (c) the tension T in the rope.

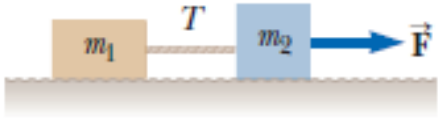
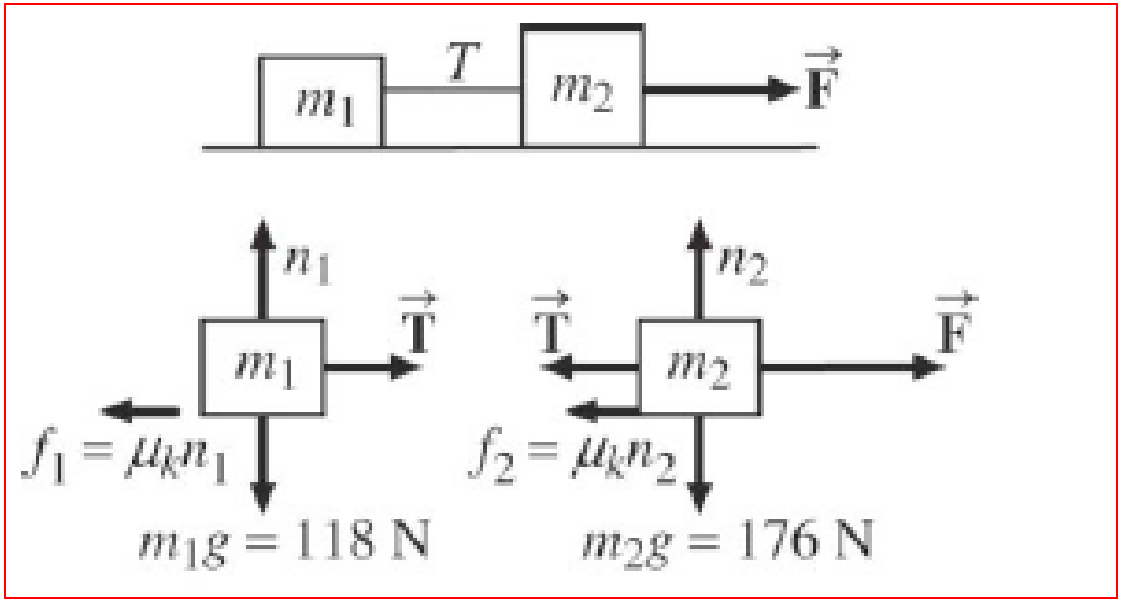


Figure P5.65

(a)



(b) We use the free-body diagrams to apply Newton's second law.

For m_1: $\sum F_x = T - f_1 = m_1 a$ or $T = m_1 a + f_1$ **[1]**

And also $\sum F_y = n_1 - m_1 g = 0$ or $n_1 = m_1 g$

Also, the definition of the coefficient of friction gives

$$f_1 = \mu n_1 = (0.100)(12.0\ \text{kg})(9.80\ \text{m/s}^2) = 11.8\ \text{N}$$

For m_2: $\sum F_x = F - T - f_2 = ma$ **[2]**

Also from the y component, $n_2 - m_2 g = 0$ or $n_2 = m_2 g$

And again $f_2 = \mu n_2 = (0.100)(18.0\ \text{kg})(9.80\ \text{m/s}^2) = 17.6\ \text{N}$

Substituting T from equation [1] into [2], we get

$$F - m_1 a - f_1 - f_2 = m_2 a \quad \text{or} \quad F - f_1 - f_2 = m_2 a + m_1 a$$

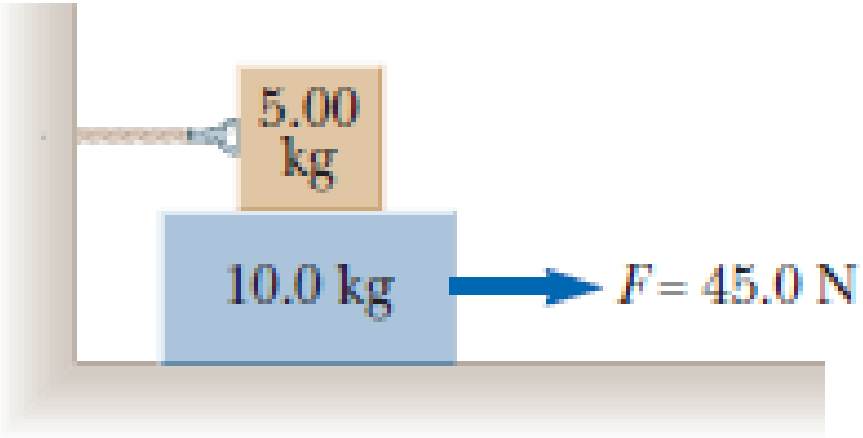
Solving for a,

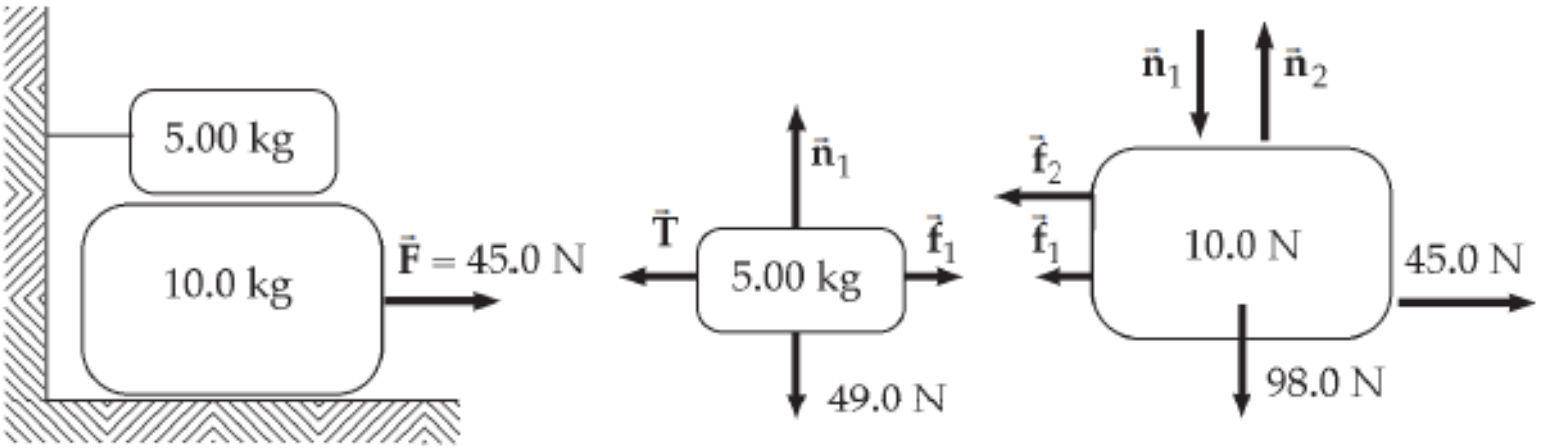
$$a = \frac{F - f_1 - f_2}{m_1 + m_2} = \frac{(68.0\ \text{N} - 11.8\ \text{N} - 17.6\ \text{N})}{(12.0\ \text{kg} + 18.0\ \text{kg})} = \boxed{1.29\ \text{m/s}^2}$$

(c) From equation [1],

$$T = m_1 a + f_1 = (12.0\ \text{kg})(1.29\ \text{m/s}^2) + 11.8\ \text{N} = \boxed{27.2\ \text{N}}$$

70. A 5.00-kg block is placed on top of a 10.0-kg block (Fig. P5.70). A horizontal force of 45.0 N is applied to the 10-kg block, and the 5.00-kg block is tied to the wall. The coefficient of kinetic friction between all moving surfaces is 0.200. (a) Draw a free-body diagram for each block and identify the action–reaction forces between the blocks. (b) Determine the tension in the string and the magnitude of the acceleration of the 10.0-kg block.





ANS. FIG. P5.70(a)

f_1 and n_1 appear in both diagrams as action-reaction pairs.

(b) For the 5.00-kg mass, Newton's second law in the y direction gives:

$$n_1 = m_1 g = (5.00\ \text{kg})(9.80\ \text{m/s}^2) = 49.0\ \text{N}$$

In the x direction,

$$f_1 - T = 0$$

$$T = f_1 = \mu mg = 0.200(5.00\ \text{kg})(9.80\ \text{m/s}^2) = \boxed{9.80\ \text{N}}$$

For the 10.0-kg mass, Newton's second law in the x direction gives:

$$45.0\ \text{N} - f_1 - f_2 = (10.0\ \text{kg})a$$

In the y direction,

$$n_2 - n_1 - 98.0\ \text{N} = 0$$

$$f_2 = \mu n_2 = \mu(n_1 + 98.0\ \text{N}) = 0.20(49.0\ \text{N} + 98.0\ \text{N}) = 29.4\ \text{N}$$

$$45.0\ \text{N} - 9.80\ \text{N} - 29.4\ \text{N} = (10.0\ \text{kg})a$$

$$a = \boxed{0.580\ \text{m/s}^2}$$